

Supplementary File S1

RQ1: Reinforcement-Learning Method-Family Characterization and Recommendations

Supporting material for *What the Survey Literature Says About Reinforcement Learning for Building Control: A Critical Synthesis*

Scope and coding scale

This supplementary file provides the survey-level evidence audit for RQ1. It reports method-family characterization and recommendation scores, evidence justifications, divergence rationales, strategy-level notes, and concise survey-level answers.

Score	Evidence-intensity interpretation
0	Not addressed
1	Mentioned
2	Discussed
3	Structuring role

Contents

Scope and coding scale	1
S1.1 Reinforcement learning for building controls: The opportunities and challenges [1]	3
S1.2 The reinforcement learning method for occupant behavior in building control: A review [2]	5
S1.3 Systematic review on deep reinforcement learning-based energy management for different building types [3]	7
S1.4 Reinforcement learning for HVAC control in intelligent buildings: A technical and conceptual review [4]	9
S1.5 A review of deep reinforcement learning for smart building energy management [5]	11
S1.6 A review of reinforcement learning for controlling building energy systems from a computer science perspective [6]	13
S1.7 Applications of Deep Reinforcement Learning for Home Energy Management Systems: A Review [7]	15

S1.8	Applications of reinforcement learning for building energy efficiency control: A review [8]	17
S1.9	Building energy management with reinforcement learning and model predictive control: A survey [9]	19
S1.10	Reinforcement learning: theory and applications in HEMS [10]	21
S1.11	A review of reinforcement learning for autonomous building energy management [11]	23
S1.12	Multi objectives reinforcement learning for smart buildings: A systematic review of algorithms, applications and future perspectives [12]	25
S1.13	Reinforcement learning for control and optimization of real buildings: Identifying and addressing implementation hurdles [13]	27
S1.14	Dynamic indoor thermal environment using reinforcement learning-based controls: Opportunities and challenges [14]	28
S1.15	A review of reinforcement learning applications to control of heating, ventilation and air conditioning systems [15]	30
S1.16	An overview of reinforcement learning-based approaches for smart home energy management systems with energy storages [16]	31
S1.17	Reinforcement learning for optimizing renewable energy utilization in buildings: A review on applications and innovations [17]	34
	References	36

S1.1 Reinforcement learning for building controls: The opportunities and challenges [1]

Table S1.1: RQ1 recoded audit table for Wang and Hong (2020). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Wang and Hong (2020)	3	2	The survey characterizes RL methods through algorithm families as one of its main review perspectives. Value-based RL is structurally dominant in the reviewed literature, while policy-gradient and actor–critic methods are discussed as rising families because explicit policy representations may support transfer learning across buildings. Model-based RL is acknowledged but treated cautiously because learning accurate models is time-consuming and expertise-dependent. Recommendation receives score 2 because the survey conditionally supports value-based RL for its practical simplicity and supports policy-gradient/actor–critic methods for transfer-learning potential, but does not present a single universally preferred family.

Table S1.2: RQ1 method-family characterization scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Wang and Hong (2020)	1	3	2	2	0

Table S1.3: RQ1 method-family recommendation scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Wang and Hong (2020)	0	2	2	2	0

Table S1.4: RQ1 divergence/rationale table for Wang and Hong (2020), after recoding.

Family	Why the survey supports it	Why the survey is cautious
Value-based	Value-based RL is presented as the dominant family in the reviewed building-control literature, especially because Q-learning-style approaches balance implementation simplicity and control performance.	Although dominant historically, value-based RL is not framed as the complete future solution for transfer, generalization, or larger continuous-control settings.
Policy-gradient / policy-based	Policy-gradient methods are discussed as increasingly popular because explicit policy representations may support transfer learning across buildings.	They are not described as the dominant current family and are presented more as a rising direction than as a universal solution.
Actor-critic	Actor-critic methods are also discussed as increasingly popular and potentially useful for transfer learning and more complex control settings.	The survey does not show actor-critic methods as clearly superior overall; support remains conditional and task-dependent.
Model-based / hybrid	Model-based RL is acknowledged as a valid method family and is compared conceptually with MPC-like approaches.	The survey is cautious because learning accurate models can be time-consuming, expertise-dependent, and not always beneficial for control performance.

Table S1.5: One-line RQ1 answer for Wang and Hong (2020), after recoding.

Survey	RQ1 answer
Wang and Hong (2020)	Wang and Hong (2020) portray value-based RL as the historically dominant family in building controls, while policy-gradient and actor-critic methods are framed as rising families because of their transfer-learning and generalization potential; model-based RL is acknowledged but treated cautiously because of modeling burden.

S1.2 The reinforcement learning method for occupant behavior in building control: A review [2]

Table S1.6: RQ1 recoded audit table for Han et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Han et al. (2021)	3	2	The survey structurally discusses value-based, policy-based, and actor–critic reinforcement-learning methods in the context of occupant behavior in building control. Value-based and tabular methods are historically dominant and are described as suitable for small and discrete problems because of simplicity, convergence, and robustness. However, the survey also states that classical tabular Q-learning becomes insufficient for stochastic and complex occupant-behavior control, especially under continuous or high-dimensional state spaces. Policy-based methods are discussed as suitable for larger or continuous problems, while actor–critic is presented as an alternative that still needs more applications. Recommendation receives score 2 because the survey gives problem-dependent guidance rather than one universal preferred method family.

Table S1.7: RQ1 method-family characterization scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Han et al. (2021)	0	3	2	2	0

Table S1.8: RQ1 method-family recommendation scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Han et al. (2021)	0	2	2	1	0

Table S1.9: RQ1 divergence/rationale table for Han et al. (2021), after recoding.

Family	Why the survey supports it	Why the survey is cautious
Value-based	Value-based methods are easy to implement, can guarantee convergence in tabular settings, are robust, and remain the most common family in reviewed occupant-behavior RL studies. They are conditionally suitable for small and discrete state/action problems.	Classical tabular Q-learning becomes insufficient for stochastic, complex, continuous, or high-dimensional occupant-behavior problems, where approximation or richer representations are needed.
Policy-gradient / policy-based	Policy-based methods are conditionally supported for large MDPs and continuous-state problems because they avoid explicit tabular value representation and can directly represent policies.	They are not framed as universally best; practical implementation and wider empirical evidence remain less developed than for value-based methods.
Actor-critic	Actor-critic methods are presented as a distinct alternative for more complex settings.	The survey explicitly treats actor-critic as less mature in this application area and states that more applications need to be developed.

Table S1.10: RQ1 strategy-level notes for Han et al. (2021), after recoding.

Survey	note_type	note
Han et al. (2021)	Training strategy	Off-policy methods are favored for complex control tasks because learning becomes more efficient when exploration and repeated real signal collection are difficult.
Han et al. (2021)	Specific algorithm guidance	SARSA may be preferred when the agent is exploring in error-insensitive systems.
Han et al. (2021)	Specific algorithm guidance	Q-learning is preferred when agents do not explore.

Table S1.11: One-line RQ1 answer for Han et al. (2021), after recoding.

Survey	RQ1 answer
Han et al. (2021)	Han et al. (2021) present a problem-dependent algorithm narrative: value-based methods remain historically dominant and suitable for small discrete problems, but classical tabular Q-learning is increasingly insufficient for stochastic and complex occupant-behavior control; policy-based methods are conditionally supported for larger or continuous problems, while actor-critic is treated as a promising but less mature alternative.

S1.3 Systematic review on deep reinforcement learning-based energy management for different building types [3]

Table S1.12: RQ1 recoded audit table for Shaqour and Hagishima (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Shaqour and Hagishima (2022)	3	3	The survey structurally characterizes DRL algorithm families across different building types using extensive algorithm tables and building-type-specific comparisons. Value-based methods such as DQN, DDQN, BDQ, and fitted Q-iteration are repeatedly visible and remain important, especially in residential studies. Actor–critic methods such as DDPG, A3C, A2C, PPO, SAC, TD3, ACKTR, and MAAC are strongly represented across residential, office, and data-center contexts. Model-based or hybrid methods such as RLMPG and model-based DRL are acknowledged but less central. Multi-agent and distributed DRL methods are structurally emphasized and explicitly recommended for future neighbourhood- and multi-building energy management. Recommendation receives score 3 because the survey gives clear future-oriented support for multi-agent/distributed DRL and also explicitly recommends SAC for future utilization in data-center HVAC.

Table S1.13: RQ1 method-family characterization scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Shaqour and Hagishima (2022)	2	3	0	3	3

Table S1.14: RQ1 method-family recommendation scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Shaqour and Hagishima (2022)	1	2	0	3	3

Table S1.15: RQ1 divergence/rationale table for Shaqour and Hagishima (2022), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	DQN and related value-based methods are repeatedly visible in the reviewed literature and remain practically important, especially in residential building studies.	The survey does not elevate value-based methods as the single best family across all building types; later and more complex contexts show diversification toward actor-critic, multi-agent, and hybrid methods.
Actor-critic	DDPG, A3C, A2C, PPO, SAC, TD3, ACKTR, and MAAC are strongly represented across building-type tables; SAC is explicitly recommended for future utilization in data-center HVAC because of its sample efficiency in noisy environments.	The survey still presents trade-offs across algorithms and building types rather than a single universally best actor-critic method.
Model-based / hybrid	Hybrid methods such as model-based DRL and RLMPc are presented as useful combinations that can leverage multiple method families.	Model-based/hybrid methods are not presented as the main overall recommendation or dominant future direction.
Multi-agent / distributed	Multi-agent and distributed approaches are explicitly framed as key future methods for neighbourhoods, multi-building settings, and coordinated energy management.	The strongest support is future-oriented and scaling-oriented rather than a universal recommendation for every building type or task.

Table S1.16: One-line RQ1 answer for Shaqour and Hagishima (2022), after recoding.

Survey	RQ1 answer
Shaqour and Hagishima (2022)	Shaqour and Hagishima (2022) present a building-type- and application-dependent algorithm narrative rather than a single dominant winner: value-based methods remain prominent through DQN/DDQN-style approaches, actor-critic methods are strongly supported through DDPG, A2C/A3C, PPO, ACKTR, SAC, TD3, and related algorithms, and the clearest future-oriented recommendation is distributed and multi-agent DRL for neighbourhood- and multi-building energy management.

S1.4 Reinforcement learning for HVAC control in intelligent buildings: A technical and conceptual review [4]

Table S1.17: RQ1 recoded audit table for Al Sayed et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Al Sayed et al. (2024)	3	2	The survey structurally characterizes reinforcement-learning algorithm families for HVAC control through a dedicated classification section and an implementation overview. Value-based methods, especially DQN, are presented as the current leading trend and are linked to discrete action-space control. Actor–critic methods are also structurally important, with DDPG and SAC identified as leading algorithms and with PPO/SAC examples showing strong HVAC-control performance. Policy-based methods are included in the theoretical taxonomy, but the paper states that no algorithms from this category have yet been implemented in BEMS. Model-based methods are less common in the reviewed literature, but are discussed as sample-efficient and potentially useful for reducing costly real-world interactions. Recommendation receives score 2 because the paper gives conditional support to value-based and actor–critic approaches as current practice, while future-oriented support is stronger for model-based RL and meta-RL as ways to improve sample efficiency, adaptation, and real-building readiness.

Table S1.18: RQ1 method-family characterization scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Al Sayed et al. (2024)	3	2	3	2	0

Table S1.19: RQ1 method-family recommendation scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Al Sayed et al. (2024)	2	0	2	2	0

Table S1.20: RQ1 divergence/rationale table for Al Sayed et al. (2024), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Value-based methods are described as the current leading trend in the HVAC RL literature, with DQN identified as the dominant algorithm. They are linked to practical use in discrete action-space control.	The paper’s future-direction logic does not simply endorse continuing with current dominant methods; it stresses that real-world deployment, adaptation, and generalization remain unresolved.
Actor-critic	Actor-critic methods form a significant part of the literature, with DDPG and SAC identified as leading algorithms. The paper also reports examples where PPO or SAC-style methods improve HVAC-control performance compared with simpler baselines.	Actor-critic methods are not presented as universally best; the paper stresses broader needs such as better generalization, safety, and real-building adaptability.
Policy-gradient / policy-based	Policy-based methods are included in the theoretical RL taxonomy and therefore are part of the method-family characterization.	The paper explicitly states that this category has not yet been implemented in BEMS, so it receives no clear practical recommendation.
Model-based / hybrid	Model-based RL is discussed as sample-efficient and potentially useful because synthetic samples or learned models can reduce costly real-world interactions.	Model-based controllers make up only a smaller portion of the reviewed literature, and accurate environment modeling remains difficult.
Meta-RL	Meta-RL is highlighted as an up-and-coming solution for improving generalization and reducing the computational cost of adaptation before real deployment.	Meta-RL is kept as a future-direction note rather than a main family column because it is discussed mainly as a generalization/adaptation strategy.

Table S1.21: RQ1 future-direction notes for Al Sayed et al. (2024), after recoding.

Survey	note_type	note
Al Sayed et al. (2024)	Future direction	Meta-RL is presented as an up-and-coming solution for improving generalization and reducing the computational cost of adaptation before real deployment.
Al Sayed et al. (2024)	Future direction	Future work should focus on real-building deployment by addressing poor diversification of exogenous state components during training.
Al Sayed et al. (2024)	Future direction	Model-based RL is presented as potentially beneficial for learning from limited interactions through synthetic samples or learned environment models.

Table S1.22: One-line RQ1 answer for Al Sayed et al. (2024), after recoding.

Survey	RQ1 answer
Al Sayed et al. (2024)	Al Sayed et al. (2024) characterize the current HVAC RL literature as led by value-based methods, especially DQN, with actor-critic methods forming a substantial second tier led by DDPG and SAC, while pure policy-based methods remain practically absent; the paper’s clearest future-oriented recommendation is to improve generalization and real-building adaptability, with model-based RL and Meta-RL highlighted as promising routes.

S1.5 A review of deep reinforcement learning for smart building energy management [5]

Table S1.23: RQ1 recoded audit table for Yu et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Yu et al. (2021)	3	3	The survey structurally classifies DRL methods for smart-building energy management and gives representative algorithms across value-based, policy-based, model-based, and multi-agent families. Value-based DRL, including DQN and its variants, is treated as a core family for discrete-action SBEM, while policy-based or actor–critic-style DRL methods such as DDPG, PPO, and A2C/A3C are discussed as representative algorithms for continuous control, stable learning, and reliable training. Model-based DRL is treated as a major family because it can improve sample complexity by learning or using environment models. Multi-agent DRL is structurally important because the paper frames multi-zone, multi-building, and microgrid-level SBEM as scalability problems that require multi-agent formulations. Recommendation receives score 3 because the paper gives strong future-oriented support to model-based DRL for sample efficiency and to multi-agent DRL for scalable practical large-building or multi-building control, while giving conditional support to value-based and policy-based methods depending on action-space structure and learning requirements.

Table S1.24: RQ1 method-family characterization scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Yu et al. (2021)	3	3	3	3	3

Table S1.25: RQ1 method-family recommendation scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Yu et al. (2021)	2	2	2	3	3

Table S1.26: RQ1 divergence/rationale table for Yu et al. (2021), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	DQN and its variants are presented as representative model-free DRL algorithms for SBEM. Experience replay can improve data efficiency, and value-based methods remain natural for discrete action-space problems.	Value-based methods are limited by discrete action support and risks such as Q-value overestimation or suboptimal policies in more complex settings.
Policy-gradient / policy-based	Policy-based and related methods are presented as representative DRL options for SBEM, especially where continuous action spaces, stable learning, or reliable training are important.	Different algorithms have different trade-offs; some may be sensitive to hyperparameters, architecture, or learning instability.
Actor-critic	Actor-critic-style algorithms such as DDPG, PPO, and A2C/A3C are discussed as representative methods for continuous-control and practical SBEM tasks.	The paper does not present one actor-critic method as universally superior; algorithm suitability depends on action-space structure and system complexity.
Model-based / hybrid	Model-based DRL is strongly supported because it can outperform model-free algorithms in sample complexity and can use learned or calibrated environment models to generate future episodes.	Accurate building-environment modeling is difficult, which limits practical applicability.
Multi-agent / distributed	Multi-agent DRL is strongly supported for scalable multi-zone, multi-building, and microgrid-level SBEM. The paper explicitly formulates large-scale SBEM as a Markov-game problem and expects more scalable multi-agent DRL algorithms for practical large buildings.	Existing multi-agent approaches are still not fully scalable for all practical large-building problems, so further algorithmic development is needed.

Table S1.27: One-line RQ1 answer for Yu et al. (2021), after recoding.

Survey	RQ1 answer
Yu et al. (2021)	Yu et al. (2021) present a capability-driven algorithm narrative rather than a single dominant winner: value-based DRL is treated as a representative discrete-action baseline, policy-based and actor-critic-style DRL are emphasized for continuous control and stable learning, model-based DRL is strongly highlighted for sample efficiency, and multi-agent DRL is positioned as increasingly important for scalable multi-zone, multi-building, and microgrid-level smart-building energy management.

S1.6 A review of reinforcement learning for controlling building energy systems from a computer science perspective [6]

Table S1.28: RQ1 recoded audit table for Weinberg et al. (2023). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Weinberg et al. (2023)	3	3	The survey is recommendation-driven rather than dominance-driven. It characterizes value-based methods such as SARSA, Q-learning, and DQN as representative algorithms, but also identifies their limitations for continuous HVAC/BES control. Policy-based and actor-critic methods such as TRPO, PPO, DDPG, and SAC are structurally important because the paper emphasizes sample efficiency, off-policy learning, and continuous-control suitability. Model-based RL is strongly emphasized as a remedy for slow data collection and low sample efficiency in building energy systems. Multi-agent RL is also structurally important because the paper organizes one of its core challenge categories around multi-agent aspects and building clusters. Recommendation receives score 3 because the paper strongly favors sample-efficient directions, especially off-policy actor-critic methods, model-based RL, transfer learning, and multi-agent RL for scalable building-cluster control.

Table S1.29: RQ1 method-family characterization scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor-critic	Model-based / hybrid	Multi-agent / distributed
Weinberg et al. (2023)	2	3	3	3	3

Table S1.30: RQ1 method-family recommendation scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor-critic	Model-based / hybrid	Multi-agent / distributed
Weinberg et al. (2023)	0	2	3	3	3

Table S1.31: RQ1 divergence/rationale table for Weinberg et al. (2023), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Value-based algorithms such as DQN, SARSA, and Q-learning are presented as standard representative RL methods and appear in the paper’s taxonomy of popular approaches.	The paper notes that Q-learning, DQN, and SARSA are unsuitable for HVAC control when the action space is large or continuous, so it does not recommend value-based methods as the general solution.
Policy-gradient / policy-based	Policy-based methods such as TRPO, PPO, DDPG, and SAC are characterized as popular RL algorithms and relevant to continuous-control settings.	The strongest recommendation is not for policy-gradient methods as a broad family, but for sample-efficient off-policy and actor–critic-style learning.
Actor–critic	Off-policy actor–critic methods are strongly supported because they can reuse collected building data and improve sample efficiency. SAC and DDPG are highlighted as especially relevant to continuous HVAC/BES control.	The paper still treats these as part of a broader sample-efficiency agenda rather than a universal solution independent of data, dynamics, or deployment context.
Model-based / hybrid	Model-based RL is strongly supported as a remedy for low sample efficiency and slow data collection, and the paper reports examples where model-based methods outperform model-free alternatives in sample efficiency or average reward.	Accurate environment modeling remains difficult, and model-based success depends on useful learned or engineered dynamics models.
Multi-agent / distributed	Multi-agent RL is structurally important because building clusters and multi-agent aspects form a core challenge category. It is treated as a key direction for scalable BES control.	Multi-agent RL is framed mainly as a future-oriented direction for clusters rather than a mature default for all BES settings.
Transfer learning	Transfer learning is highlighted as a prominent strategy for efficient learning in building clusters, either by transferring control policies or plant models between buildings.	Transfer learning is kept as a strategy note rather than a main family column because the main family heatmap focuses on algorithm classes.

Table S1.32: RQ1 strategy-level notes for Weinberg et al. (2023), after recoding.

Survey	note_type	note
Weinberg et al. (2023)	Training strategy	Off-policy algorithms are described as attractive for building energy systems because slow sensor sampling makes data expensive, and off-policy methods can reuse all collected data throughout learning.
Weinberg et al. (2023)	Algorithm comparison	DDPG is reported as converging much faster than DQN for continuous-action HVAC control.
Weinberg et al. (2023)	Algorithm comparison	Off-policy SAC is reported as converging much faster than on-policy counterparts in a simulated data-center setting, with SAC framed as a likely key method for next-generation HVAC control systems.
Weinberg et al. (2023)	Future direction	Transfer learning is highlighted as a prominent approach for efficient learning in building clusters, either by transferring control policies or plant models between buildings.

Table S1.33: One-line RQ1 answer for Weinberg et al. (2023), after recoding.

Survey	RQ1 answer
Weinberg et al. (2023)	Weinberg et al. (2023) present a recommendation-driven rather than dominance-driven algorithm narrative: instead of naming one classical RL family as the overall winner, they favor sample-efficient approaches for building energy systems, especially off-policy actor–critic learning, model-based RL, transfer learning, and multi-agent RL for scalable control in building clusters.

S1.7 Applications of Deep Reinforcement Learning for Home Energy Management Systems: A Review [7]

Table S1.34: RQ1 recoded audit table for Latoń et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Latoń et al. (2024)	3	3	The survey structurally characterizes RL and DRL methods for HEMS across multiple application areas, including demand response, load scheduling, renewable integration, energy storage, electric vehicles, prosumer systems, and distributed home-energy automation. Value-based methods such as Q-learning, DQN, Double DQN, and Dueling DQN are repeatedly represented across HEMS use cases. Actor–critic and modern DRL methods such as PPO, DDPG, A2C, ACKTR, SAC, TD3, and related continuous-control approaches are also widely represented across application groups. Multi-agent and distributed reinforcement learning are structurally important because the review connects them to decentralized control, privacy preservation, scalability, smart-grid interaction, local microgrids, and prosumer-level coordination. Recommendation receives score 3 because the clearest future-oriented direction is toward scalable, privacy-aware, distributed and multi-agent RL/DRL, supported by transfer learning, stronger interoperability, and more real-world deployable HEMS architectures.

Table S1.35: RQ1 method-family characterization scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor–critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Latoń et al. (2024)	3	0	3	3	0	3

Table S1.36: RQ1 method-family recommendation scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor-critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Latoń et al. (2024)	2	0	2	2	0	3

Table S1.37: RQ1 divergence/rationale table for Latoń et al. (2024), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Q-learning and DQN-family methods appear across several HEMS application groups, showing that they remain practically important in demand response, scheduling, renewable-energy integration, storage, and electric-vehicle applications.	The review presents HEMS control as application-dependent and does not elevate value-based methods as the single leading family across all tasks, especially where continuous control, coordination, privacy, and scalability matter.
Actor-critic / modern DRL	PPO, DDPG, ACKTR, A2C, SAC, TD3, and related methods appear across multiple HEMS application areas, especially where richer DRL implementations, scheduling, continuous control, and adaptive optimization are needed.	The paper does not rank these methods as universally best; they are treated as part of a broad, application-dependent algorithm mix.
Deep RL	DRL is central to the survey’s HEMS narrative because it enables more adaptive and complex decision-making than simpler tabular approaches in modern smart-home energy systems.	DRL remains constrained by real-world validation gaps, data requirements, scalability, privacy/security, and interoperability challenges.
Multi-agent / distributed	MARL and distributed control are described as promising for decentralized, privacy-preserving, scalable energy management in smart homes, prosumer systems, local microgrids, and smart-grid-integrated HEMS.	The paper also notes major practical challenges, including limited transferability across heterogeneous buildings, costly retraining, and the need for adaptive learning frameworks.

Table S1.38: RQ1 future-direction notes for Latoń et al. (2024), after recoding.

Survey	note_type	note
Latoń et al. (2024)	Future direction	The review highlights a strong need for more real-world implementations and case studies, since most existing HEMS RL/DRL studies remain simulation-based.
Latoń et al. (2024)	Future direction	Transfer learning is presented as a promising way to improve the adaptability of RL and MARL across buildings and systems with limited data.
Latoń et al. (2024)	Future direction	Security, privacy, scalability, interoperability, and communication standards are emphasized as major challenges that future RL/DRL-based HEMS research must address.

Table S1.39: One-line RQ1 answer for Latoń et al. (2024), after recoding.

Survey	RQ1 answer
Latoń et al. (2024)	Latoń et al. (2024) present a mixed, application-dependent DRL narrative for HEMS rather than a single dominant family: both value-based and actor-critic/modern DRL methods are broadly represented across use cases, while the clearest future-oriented recommendation is toward multi-agent and distributed RL, transfer learning, and more scalable, secure, privacy-aware, and real-world deployable energy-management systems.

S1.8 Applications of reinforcement learning for building energy efficiency control: A review [8]

Table S1.40: RQ1 recoded audit table for Fu et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Fu et al. (2022)	3	3	The survey provides one of the clearest algorithm-family narratives in the corpus. Value-based algorithms are structurally dominant in the reviewed literature, with Q-learning described as the most classical and most commonly used method, and DQN-like variants extending value-based RL to larger or continuous state spaces with discrete actions. Policy-search methods are explicitly discussed but are less common and less preferred because of policy-evaluation cost. Actor–critic methods are structurally important because they combine value-based and policy-based advantages and are explicitly prioritized for practical problems with large or continuous action spaces. Multi-agent reinforcement learning is also strongly future-oriented because the paper recommends it for complex control scales, multiple control objects, building clusters, and multi-building control. Recommendation receives score 3 because actor–critic and MARL are explicitly recommended for complex, continuous, and multi-object building-control problems, while value-based methods remain conditionally recommended for simpler small/discrete settings.

Table S1.41: RQ1 method-family characterization scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor–critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Fu et al. (2022)	3	2	3	3	0	3

Table S1.42: RQ1 method-family recommendation scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor–critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Fu et al. (2022)	2	1	3	3	0	3

Table S1.43: RQ1 divergence/rationale table for Fu et al. (2022), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Value-based methods dominate the reviewed literature. Q-learning is simple, easy to deploy, and preferred for small/discrete problems, while DQN-like methods extend the family to large or continuous state spaces with discrete actions.	Value-based methods are weak for continuous action spaces, and discretization can lead to suboptimal control policies compared with policy-based or actor-critic methods.
Policy-search / policy-based	Policy-search methods can handle large or continuous action spaces, and PPO is presented as relatively stable.	The survey says policy-search methods are less used, more costly in policy evaluation, and not the first choice compared with actor-critic methods.
Actor-critic	Actor-critic methods combine the advantages of value-based and policy-based approaches, are more efficient for practical applications, and are explicitly prioritized for complex control problems with large or continuous action spaces. DDPG is highlighted for continuous actions, while A2C/A3C are linked to faster training.	Actor-critic methods are more difficult to implement than simpler value-based methods and are not presented as necessary for small or simple discrete problems.
Deep RL	Deep reinforcement learning is presented as more appropriate than classical RL for complex building-control problems.	DRL still inherits practical limitations such as training time, safety, and implementation difficulty.
Multi-agent / distributed	Multi-agent reinforcement learning is explicitly recommended for building clusters and multiple control objects, and is reported to have better efficiency and performance than single-agent approaches in complex systems.	MARL is framed mainly as a future direction needing more recent research and application before it becomes a mature default.

Table S1.44: RQ1 future-direction notes for Fu et al. (2022), after recoding.

Survey	note_type	note
Fu et al. (2022)	Future direction	Deep reinforcement learning is explicitly presented as more appropriate than existing RL methods for solving more complex building-control problems.
Fu et al. (2022)	Future direction	Multi-agent reinforcement learning is highlighted as having better control efficiency and performance than single-agent RL for complex building-control problems.
Fu et al. (2022)	Future direction	Reinforcement learning combined with occupant behavior is identified as a promising future research direction.

Table S1.45: One-line RQ1 answer for Fu et al. (2022), after recoding.

Survey	RQ1 answer
Fu et al. (2022)	Fu et al. (2022) present one of the clearest algorithm-narrative shifts in the corpus: value-based RL still dominates the reviewed intelligent-building literature, especially through Q-learning, but actor-critic methods are explicitly prioritized for large or continuous action spaces, deep RL is favored for more complex control problems, and multi-agent RL is recommended for complex multi-object and building-cluster control.

S1.9 Building energy management with reinforcement learning and model predictive control: A survey [9]

Table S1.46: RQ1 recoded audit table for Zhang et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Zhang et al. (2022)	3	3	The survey frames RQ1 primarily as a control-design and method-selection problem rather than a simple algorithm-family ranking. Classical value-based methods such as Q-learning and SARSA are discussed as part of the RL foundation for BEMS, while actor-critic methods are included as relevant advanced RL approaches. Deep RL is structurally important because it extends RL to complex and high-dimensional BEMS settings. The strongest characterization and recommendation are for model-based and hybrid RL-MPC approaches: the paper argues that MPC should be preferred when transition dynamics are well understood, while RL becomes attractive when environments are difficult to model, and hybrid/model-based RL can combine RL adaptability with MPC-style planning, data efficiency, safety, reliability, and robustness. Recommendation receives score 3 because RL+MPC/model-based RL is the clearest future-oriented direction in the survey.

Table S1.47: RQ1 method-family characterization scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor-critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Zhang et al. (2022)	2	0	2	3	3	0

Table S1.48: RQ1 method-family recommendation scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor-critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Zhang et al. (2022)	0	0	1	2	3	0

Table S1.49: RQ1 divergence/rationale table for Zhang et al. (2022), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Value-based methods provide the classical RL foundation in BEMS, with Q-learning and SARSA serving as representative baseline methods under unknown dynamics.	Classical value-based methods are limited in high-dimensional and continuous-control settings, so the paper does not present them as the main practical future direction.
Actor-critic	Actor-critic methods are included as a core RL family and are discussed in BEMS applications, indicating relevance for more advanced control settings.	Actor-critic is not singled out as the dominant or most recommended family; support is weaker than for hybrid/model-based approaches.
Deep RL	Deep RL is presented as necessary for extending RL to more complex and high-dimensional BEMS problems and as an important development beyond classical RL.	Deep RL remains constrained by data-efficiency, safety, robustness, and generalization challenges, which the paper treats as major practical barriers.
Model-based / hybrid RL	Model-based and hybrid RL-MPC approaches receive the strongest support because they can combine RL adaptability with MPC-style planning and improve data efficiency, safety, reliability, and robustness.	These approaches still depend on useful models, predictive structure, or simulator support, so their applicability is conditional on model availability and problem setup.

Table S1.50: RQ1 future-direction notes for Zhang et al. (2022), after recoding.

Survey	note_type	note
Zhang et al. (2022)	Recommendation logic	The paper argues that if transition dynamics are well understood, MPC should be preferred; if the environment is difficult to model, RL becomes attractive.
Zhang et al. (2022)	Future direction	The survey highlights hybrid RL + MPC approaches as a promising compromise between reliability and practicality in real-time BEMS.
Zhang et al. (2022)	Future direction	Data efficiency, safety, robustness, and generalization are emphasized as central reasons to move toward model-based and hybrid RL formulations.

Table S1.51: One-line RQ1 answer for Zhang et al. (2022), after recoding.

Survey	RQ1 answer
Zhang et al. (2022)	Zhang et al. (2022) frame RQ1 primarily as a control-design choice rather than a family-dominance question: instead of naming one RL family as state of the art, the survey recommends choosing among classical RL, deep RL, and especially hybrid/model-based RL according to model availability, data efficiency, safety, reliability, and robustness requirements, with the strongest support going to RL+MPC and model-based RL directions.

S1.10 Reinforcement learning: theory and applications in HEMS [10]

Table S1.52: RQ1 recoded audit table for Al-Ani and Das (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Al-Ani and Das (2022)	3	3	The survey structurally characterizes deep reinforcement-learning families for HEMS and shows that the field had already diversified beyond classical RL. Deep value-based methods, especially DQN and DDQN, are a major discrete-action strand in the reviewed HEMS literature. Policy-based methods such as PPO and TRPO are present but thinly represented. Actor–critic methods are the strongest family in the survey’s deep-learning landscape, with DDPG, SAC, A2C, A3C, and TD3 listed as representative algorithms and actor–critic described as preferred over deep policy learning. Recommendation receives score 3 because the clearest method-family preference is toward actor–critic approaches, while value-based methods remain conditionally important for discrete-action HEMS settings.

Table S1.53: RQ1 method-family characterization scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-based	Actor–critic
Al-Ani and Das (2022)	3	2	3

Table S1.54: RQ1 method-family recommendation scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-based	Actor–critic
Al-Ani and Das (2022)	2	0	3

Table S1.55: RQ1 divergence/rationale table for Al-Ani and Das (2022), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	DQN is the most-used deep algorithm in the reviewed HEMS literature, and DDQN is also popular, making deep value-based methods the standard discrete-action deep-RL option in this survey.	The paper does not elevate value-based methods as the overall preferred deep-learning family; its stronger preference shifts toward actor-critic methods.
Policy-based	PPO and TRPO appear in the HEMS literature, so the family is not absent and receives method-family characterization.	The paper indicates that only PPO and TRPO have been used so far in HEMS and treats deep policy learning as much less preferred than actor-critic methods.
Actor-critic	Actor-critic is presented as the preferred deep-learning approach relative to deep policy learning and as the largest deep-learning family in the survey, with broad coverage through DDPG, SAC, A2C, A3C, and TD3.	The paper's recommendation is comparative rather than universal; it prefers actor-critic over deep policy learning, but does not claim that it solves all HEMS control problems without qualification.

Table S1.56: RQ1 future-direction notes for Al-Ani and Das (2022), after recoding.

Survey	note_type	note
Al-Ani and Das (2022)	Family-use pattern	The paper shows that HEMS deep RL had already diversified beyond classical RL, but the strongest concentration was in DQN/DDQN and actor-critic methods rather than in pure deep policy learning.
Al-Ani and Das (2022)	Recommendation logic	The paper's clearest recommendation is comparative: actor-critic is preferred over deep policy learning in the reviewed HEMS literature.
Al-Ani and Das (2022)	Scope note	The review is HEMS-specific, so its RQ1 contribution is strongest for home energy management rather than the entire building-control literature.

Table S1.57: One-line RQ1 answer for Al-Ani and Das (2022), after recoding.

Survey	RQ1 answer
Al-Ani and Das (2022)	Al-Ani and Das (2022) show that HEMS-focused deep RL had already moved beyond simple value-based methods into a mixed deep-RL landscape, but with a clear hierarchy: DQN/DDQN remain the main deep value-based strand, pure deep policy learning is still very limited, and actor-critic emerges as the preferred deep-learning family in practice.

S1.11 A review of reinforcement learning for autonomous building energy management [11]

Table S1.58: RQ1 recoded audit table for Mason and Grijalva (2019). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Mason and Grijalva (2019)	3	3	The survey captures an early-stage reinforcement-learning narrative in building energy management. Value-based methods, especially Q-learning and its variants, are structurally dominant in the reviewed HVAC and home-energy-management applications. Actor–critic methods are present through several applications, including HVAC, water heaters, and HEMS, but they are not treated as the dominant or preferred family. Multi-agent and distributed reinforcement learning are structurally important because the paper discusses them in smart-home, community, and grid-related settings and explicitly recommends scaling multi-agent RL to larger communities of homes. Recommendation receives score 3 because the clearest future-oriented family-level recommendation is toward larger-scale multi-agent RL, while value-based methods remain conditionally supported as the historical practical workhorse.

Table S1.59: RQ1 method-family characterization scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Value-based	Actor–critic	Multi-agent / distributed
Mason and Grijalva (2019)	3	2	3

Table S1.60: RQ1 method-family recommendation scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Value-based	Actor–critic	Multi-agent / distributed
Mason and Grijalva (2019)	2	0	3

Table S1.61: RQ1 divergence/rationale table for Mason and Grijalva (2019), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Q-learning is repeatedly described as the most common method in HVAC and home energy management, and many reviewed studies report practical savings with Q-learning and its variants.	The future-directions section shifts attention away from purely classical RL toward deeper function approximation and more complex methods for larger state-action spaces.
Actor-critic	Actor-critic appears in several successful applications, including HVAC, water heaters, and HEMS, and is part of the broader reviewed algorithm mix.	The paper does not elevate actor-critic as the dominant or preferred family across the survey.
Multi-agent / distributed	Multi-agent RL is described as commonly used for smart-home/community and grid-integration problems, and the future-directions section explicitly recommends scaling multi-agent RL to larger communities of homes.	It is mainly associated with multi-home and distributed-control settings, not presented as the default solution for all building-energy-management tasks.

Table S1.62: RQ1 future-direction notes for Mason and Grijalva (2019), after recoding.

Survey	note_type	note
Mason and Grijalva (2019)	Future direction	The paper identifies a strong move toward deep RL methods for larger state-action spaces and more complex control tasks.
Mason and Grijalva (2019)	Future direction	The paper explicitly proposes future work on multi-task RL, transfer learning, multi-objective RL, and meta-RL for building energy management.
Mason and Grijalva (2019)	Future direction	The paper recommends extending multi-agent RL to larger communities of homes and studying scalability and cooperation in decentralized settings.

Table S1.63: One-line RQ1 answer for Mason and Grijalva (2019), after recoding.

Survey	RQ1 answer
Mason and Grijalva (2019)	Mason and Grijalva (2019) capture an early-stage reinforcement-learning narrative in building energy management: classical value-based RL, especially Q-learning, still dominates practical applications, while future work is expected to move toward deeper function-approximation methods and larger-scale multi-agent control.

S1.12 Multi objectives reinforcement learning for smart buildings: A systematic review of algorithms, applications and future perspectives [12]

Table S1.64: RQ1 recoded audit table for Huynh et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Huynh et al. (2025)	3	3	This is a specialized MORL-focused survey, so RQ1 should be coded around multi-objective reinforcement-learning design rather than only standard single-objective RL families. Traditional RL is discussed as an important baseline, but is treated as insufficient for smart-building settings with conflicting objectives and changing priorities. MORL is the structuring method family of the survey because the paper centers vector-valued rewards, MOMDPs, Pareto reasoning, scalarization, hybrid MORL, and adaptive control under multiple objectives. Single-policy strategies are structurally discussed through weighted sums, scalarization, constraint-based methods, and SER-style approaches, while multi-policy and adaptive strategies are structurally important because the paper emphasizes Pareto-front methods, preference-independent methods, adaptive policies, occupant feedback, and flexibility under changing priorities. Recommendation receives score 3 because the paper clearly recommends MORL over traditional RL for conflicting smart-building objectives and gives the strongest future-oriented support to adaptive and multi-policy approaches.

Table S1.65: RQ1 method-family characterization scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Traditional RL	MORL	Single-policy	Multi-policy / adaptive
Huynh et al. (2025)	2	3	3	3

Table S1.66: RQ1 method-family recommendation scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Traditional RL	MORL	Single-policy	Multi-policy / adaptive
Huynh et al. (2025)	0	3	2	3

Table S1.67: RQ1 divergence/rationale table for Huynh et al. (2025), after recoding.

Category	Why the survey supports it	Why the survey is cautious / limited
Traditional RL	Traditional RL provides the baseline RL framing and can be extended through reward shaping or scalarization.	The paper argues that single-objective RL is insufficient for smart buildings with multiple conflicting goals, occupant preferences, and changing priorities.
MORL	MORL is better suited to balancing energy, cost, comfort, indoor air quality, and other competing objectives. It improves adaptability and supports more effective decision-making in complex smart-building environments.	MORL is more complex and computationally demanding, which may hinder real-time implementation in large-scale buildings.
Single-policy	Single-policy methods are useful when preferences are known and fixed, and they provide simpler implementations through scalarization, weighted sums, or constraints.	They are less flexible because they produce a single optimized outcome and cannot represent a broader set of trade-offs under changing preferences.
Multi-policy / adaptive	Multi-policy and adaptive approaches are better suited to dynamic environments and changing priorities. They support Pareto-front reasoning, preference-independent decision-making, occupant feedback, and more flexible control.	They are harder to compute, more complex to manage, and depend on robust preference handling and deployment-ready control logic.

Table S1.68: RQ1 objective-frequency notes for Huynh et al. (2025).

Objective	Frequency
Energy consumption	33
Energy cost	33
Thermal comfort	33
User comfort	18
Indoor air quality	6
Peak shaving	5
Indoor temperature	5

Table S1.69: One-line RQ1 answer for Huynh et al. (2025), after recoding.

Survey	RQ1 answer
Huynh et al. (2025)	Huynh et al. (2025) show a late-stage shift in RL survey discourse: rather than asking only which standard single-objective RL family is best, the survey argues that smart-building control increasingly requires MORL frameworks that can balance conflicting objectives through adaptive and often multi-policy strategies, with growing emphasis on occupant feedback, flexibility, and real-world deployability.

S1.13 Reinforcement learning for control and optimization of real buildings: Identifying and addressing implementation hurdles [13]

Table S1.70: RQ1 inclusion-status table for Kannari et al. (2025).

Survey	Include in RQ1 family heatmap?	Use in RQ1 qualitative discussion?	Reason
Kannari et al. (2025)	No	Yes	The paper focuses on real-building implementation challenges, digital twins, safety, scalability, transparency, privacy, governance, and ethics rather than comparing or recommending RL method families such as value-based, policy-based, actor-critic, model-based, offline, or multi-agent RL.

Table S1.71: RQ1 qualitative contribution table for Kannari et al. (2025).

Survey	year	What it contributes to RQ1	What it does not contribute
Kannari et al. (2025)	2025	Shows that later review discourse shifts away from algorithm-family ranking toward deployment-oriented evaluation criteria, especially safety, scalability, digital-twin support, data sharing, transparency, privacy, governance, and ethics.	Does not provide a family-level ranking or recommendation among value-based, policy-based, actor-critic, offline RL, model-based RL, multi-agent RL, or other standard RL families.

Table S1.72: RQ1 rationale-note table for Kannari et al. (2025).

Survey	note_type	note
Kannari et al. (2025)	Rationale shift	The paper shows that recommendation logic has shifted toward safety, robustness, and long-term reliability of adaptive solutions in real buildings.
Kannari et al. (2025)	Rationale shift	The paper emphasizes scalability and wider practical applicability across building contexts as major criteria for judging RL-based control.
Kannari et al. (2025)	Rationale shift	The paper treats digital twins as important support for RL development, pretraining, deployment, and operation in real buildings.
Kannari et al. (2025)	Rationale shift	The paper highlights data sharing, transparency, privacy, ethics, governance, accountability, and human oversight as increasingly important criteria in real-building AI/RL deployment.

Table S1.73: One-line RQ1 note for Kannari et al. (2025).	
Survey	RQ1 note
Kannari et al. (2025)	Kannari et al. (2025) is not a core source for RQ1 method-family coding, but it is useful for showing that later review discourse evaluates RL control increasingly through deployment-oriented rationales such as safety, scalability, digital-twin support, data sharing, transparency, privacy, governance, and ethics rather than through algorithm-family ranking alone.

S1.14

Dynamic indoor thermal environment using reinforcement learning-based controls: Opportunities and challenges [14]

Table S1.74: RQ1 recoded audit table for Chatterjee and Khovalyg (2023). Scores use the shared 0–3 evidence-intensity scale.			
Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Chatterjee and Khovalyg (2023)	3	3	The survey structurally characterizes RL algorithms for HVAC and dynamic indoor-environment control. Value-based methods, especially Q-learning and DQN-style extensions, are presented as the historical workhorse of RL-based HVAC control, with Q-learning described as the most explored approach in the reviewed literature. Actor–critic and advanced DRL methods are also structurally important because the paper links DDPG, A3C, TD3, SAC, PPO, TRPO, and related methods to continuous-control capability, high-dimensional control, sample efficiency, stability, and dynamic indoor-environment applications. Recommendation receives score 3 because the paper’s strongest method-family recommendation shifts toward continuous-control actor–critic DRL, especially DDPG and A3C, with TD3 and SAC discussed as stronger alternatives when stability, sample efficiency, or policy expressiveness are critical.

Table S1.75: RQ1 method-family characterization scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Value-based	Actor–critic / advanced DRL
Chatterjee and Khovalyg (2023)	3	3

Table S1.76: RQ1 method-family recommendation scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Value-based	Actor-critic / advanced DRL
Chatterjee and Khovalyg (2023)	2	3

Table S1.77: RQ1 divergence/rationale table for Chatterjee and Khovalyg (2023), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Q-learning is the most explored algorithm in the reviewed HVAC literature and remains the main historical workhorse. DQN-style extensions also appear repeatedly across the surveyed studies.	Q-learning is practical mainly for simpler or minimal environments, and DQN-style methods still face action-space and discretization limits that motivate a move toward continuous-control actor-critic methods.
Actor-critic / advanced DRL	DDPG is emphasized as a strong choice for dynamic indoor-environment control because it supports continuous action spaces, stabilizes learning through the actor-critic structure, uses off-policy learning, and can operate in high-dimensional settings. A3C, TD3, and SAC are also highlighted as strong alternatives when sample efficiency, stability, or multi-modal policy representation are especially important.	These methods can be more complex, more data-intensive, slower to train, or more computationally expensive than simpler value-based approaches, so the paper presents the choice as application-dependent rather than universally settled.

Table S1.78: RQ1 future-direction notes for Chatterjee and Khovalyg (2023), after recoding.

Survey	note_type	note
Chatterjee and Khovalyg (2023)	Specialization note	The paper is specialized to dynamic indoor-environment control with triple objectives: energy, comfort, and health.
Chatterjee and Khovalyg (2023)	Future direction	The paper argues that more studies should bring dynamic indoor-environment design and RL controls under the same framework.
Chatterjee and Khovalyg (2023)	Deployment gap	The paper notes that RL controllers are still largely in the research-and-development stage, with only a minority of reviewed controllers implemented and tested in real buildings.
Chatterjee and Khovalyg (2023)	Barrier note	The major obstacles highlighted are time-consuming and data-intensive training, control security, transfer/generalization, and the need for an open, interoperable virtual testbed.

Table S1.79: One-line RQ1 answer for Chatterjee and Khovalyg (2023), after recoding.

Survey	RQ1 answer
Chatterjee and Khovalyg (2023)	Chatterjee and Khovalyg (2023) present a specialized algorithm narrative for dynamic indoor-environment control: Q-learning remains the most explored historical HVAC-control method, but the recommendation logic shifts toward continuous-control actor-critic DRL, with DDPG presented as the primary candidate and A3C, TD3, and SAC highlighted as strong alternatives when stability, sample efficiency, or policy expressiveness are critical.

S1.15 A review of reinforcement learning applications to control of heating, ventilation and air conditioning systems [15]

Table S1.80: RQ1 inclusion-status table for Sierla et al. (2022).

Survey	Include in RQ1 family heatmap?	Use in RQ1 qualitative discussion?	Reason
Sierla et al. (2022)	No	Yes	The paper focuses on how reinforcement-learning actions are formulated for HVAC control, including action abstraction, actuator commands, setpoints, single-zone and multi-zone applicability, and indoor-environment targets such as temperature, humidity, and air quality. It does not provide a family-level ranking or recommendation among value-based, policy-based, actor-critic, model-based, offline, or multi-agent reinforcement-learning families.

Table S1.81: RQ1 qualitative contribution table for Sierla et al. (2022).

Survey	year	What it contributes to RQ1	What it does not contribute
Sierla et al. (2022)	2022	Shows that method recommendation cannot be separated from control formulation. The paper shifts attention from asking which RL family is best toward asking whether the RL action representation can be mapped meaningfully to real HVAC actions, actuator commands, setpoint changes, supervisory control layers, and indoor-environment objectives.	Does not provide a family-level ranking or recommendation among standard RL algorithm families such as value-based, policy-gradient, actor-critic, model-based, offline, or multi-agent RL.

Table S1.82: RQ1 rationale-note table for Sierla et al. (2022).

Survey	note_type	note
Sierla et al. (2022)	Rationale shift	The paper shows that later RL-HVAC reviews increasingly evaluate methods through deployable action formulation rather than algorithm family alone.
Sierla et al. (2022)	Rationale shift	The paper emphasizes whether actions correspond to actuator commands, setpoint adjustments, supervisory decisions, or planning-level control.
Sierla et al. (2022)	Rationale shift	The paper highlights that action abstraction affects whether RL control is physically meaningful, scalable to multi-zone buildings, and compatible with real building automation systems.
Sierla et al. (2022)	Rationale shift	The paper is therefore useful for qualitative RQ1 synthesis, even though it should not be included in the main method-family heatmap.

Table S1.83: One-line RQ1 note for Sierla et al. (2022).

Survey	RQ1 note
Sierla et al. (2022)	Sierla et al. (2022) is not a core source for RQ1 method-family coding, but it is useful for showing that later review discourse evaluates RL methods through formulation-level criteria, especially whether actions can be mapped to realistic HVAC setpoints, actuators, control layers, and indoor-environment targets.

S1.16 An overview of reinforcement learning-based approaches for smart home energy management systems with energy storages [16]

Table S1.84: RQ1 recoded audit table for Pinthurat et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Pinthurat et al. (2024)	3	3	The survey structurally characterizes reinforcement-learning methods for smart-home energy management systems with energy storage. Value-based methods such as Q-learning, DQN, and DDQN are central to the reviewed single-agent RL literature. Policy-gradient or policy-based methods such as PPO are discussed, but are less central than value-based and actor–critic methods. Actor–critic and modern DRL methods such as DDPG, TD3, and SAC are structurally visible because they support more complex and continuous control settings. Model-based and hybrid RL are discussed through the model-free/model-based distinction and through the paper’s future emphasis on hybrid RL algorithms. Multi-agent and distributed RL are structurally important because the paper explicitly separates single-agent and multi-agent RL and treats multi-agent methods as promising for coordinated smart homes, storage systems, and multiple controllable devices. Recommendation receives score 3 because the clearest future-oriented recommendations are toward hybrid RL and multi-agent RL for scalable, coordinated, and practically deployable SHEMS with storage.

Table S1.85: RQ1 method-family characterization scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic / modern DRL	Model-based / hybrid	Multi-agent / distributed
Pinthurat et al. (2024)	3	2	3	2	3

Table S1.86: RQ1 method-family recommendation scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic / modern DRL	Model-based / hybrid	Multi-agent / distributed
Pinthurat et al. (2024)	2	1	2	3	3

Table S1.87: RQ1 divergence/rationale table for Pinthurat et al. (2024), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Q-learning, DQN, and DDQN are structurally represented in the single-agent RL literature for SHEMS with storage and remain important practical methods for discrete decision settings.	They are less suitable as the main future direction when control requires continuous decisions, coordination across devices, storage management, and adaptation to changing user, weather, and grid conditions.
Policy-gradient / policy-based	PPO and related policy-based methods are included in the algorithm review and are relevant to more flexible policy learning.	They are not presented as the dominant or clearest future family in the paper; support remains weaker than for actor–critic, hybrid, and multi-agent methods.
Actor–critic / modern DRL	DDPG, TD3, SAC, and related modern DRL methods are relevant for more complex and continuous-control SHEMS settings, especially when storage and flexible devices are included.	The paper does not make actor–critic alone the strongest future recommendation; it places stronger future emphasis on hybrid and multi-agent RL.
Model-based / hybrid	Hybrid RL is explicitly highlighted as a promising future direction because combining complementary learning or control mechanisms can improve robustness, scalability, and practical applicability.	The paper does not treat model-based/hybrid methods as the dominant current family in the reviewed literature.
Multi-agent / distributed	Multi-agent RL is strongly supported for coordinated smart-home communities, multiple energy resources, storage systems, and distributed controllable devices. It is one of the clearest future-oriented directions.	Multi-agent methods add coordination, communication, scalability, and privacy/security challenges, so the recommendation is future-oriented rather than a settled universal default.

Table S1.88: RQ1 future-direction notes for Pinthurat et al. (2024), after recoding.

Survey	note_type	note
Pinthurat et al. (2024)	Future direction	Hybrid RL algorithms are identified as a promising future direction for RL-based smart-home energy management with storage.
Pinthurat et al. (2024)	Future direction	Multi-agent RL is identified as a major future direction for coordinating multiple devices, storage systems, and smart-home/community-level energy management.
Pinthurat et al. (2024)	Future direction	Future RL methods should better support demand-side management, demand response, user preferences, privacy/security, and real-world testing.

Table S1.89: One-line RQ1 answer for Pinthurat et al. (2024), after recoding.

Survey	RQ1 answer
Pinthurat et al. (2024)	Pinthurat et al. (2024) characterize RL for smart-home energy management with storage through both single-agent and multi-agent algorithm families: value-based and actor–critic methods remain structurally important, while the strongest future-oriented recommendations point toward hybrid RL and multi-agent RL for scalable, coordinated, storage-aware, and real-world deployable SHEMS.

S1.17 Reinforcement learning for optimizing renewable energy utilization in buildings: A review on applications and innovations [17]

Table S1.90: RQ1 recoded audit table for Michailidis et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	Method-family characterization	Method-family recommendation	Recoded evidence justification
Michailidis et al. (2025)	3	3	The survey structurally classifies reinforcement-learning studies for renewable-energy-integrated BEMS by algorithmic approach, agent framework, value-action network, reward design, baseline controls, RES type, BEMS type, and building typology. Value-based methods such as Q-learning, DQN, DDQN, and D3QN are repeatedly visible in the reviewed literature. Policy-gradient or policy-based methods are discussed but are less prevalent. Actor–critic methods such as TD3, DDPG, and SAC are strongly characterized because they are suitable for continuous, high-dimensional, and adaptive RES-integrated building control. Model-based and hybrid RL are structurally important because hybrid RL is described as an emerging and rapidly growing niche that combines RL with methods such as fuzzy logic, ANN, MPC, and RBC. Multi-agent and distributed RL are also structurally important because the paper evaluates the role of multi-agent and hybrid methods in improving adaptability, coordination, and occupant comfort across RES-integrated BEMS. Recommendation receives score 3 because the strongest future-oriented directions are actor–critic, hybrid/model-based, and multi-agent approaches for scalable, safe, transferable, and coordinated RES-integrated building control.

Table S1.91: RQ1 method-family characterization scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Michailidis et al. (2025)	3	2	3	3	3

Table S1.92: RQ1 method-family recommendation scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Michailidis et al. (2025)	2	2	3	3	3

Table S1.93: RQ1 divergence/rationale table for Michailidis et al. (2025), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Q-learning, DQN, DDQN, D3QN, and other value-action-network methods are repeatedly tabulated across RES-integrated BEMS studies, showing that value-based RL remains important in the historical literature.	Value-based methods are not the strongest future direction for high-dimensional, continuous, multi-objective, and RES-integrated building-control problems.
Policy-gradient / policy-based	Policy-based methods are discussed as useful for continuous and stochastic control settings and as part of the broader RL algorithm landscape.	The paper treats this family as lower-prevalence and less central than actor-critic, hybrid, and multi-agent approaches.
Actor-critic	Actor-critic methods such as TD3, DDPG, and SAC are strongly supported for continuous, high-dimensional, adaptive, and RES-integrated BEMS control.	Their effectiveness depends on problem formulation, training stability, and deployment conditions, so they are not presented as a complete solution by themselves.
Model-based / hybrid	Hybrid RL is presented as an emerging and rapidly growing direction that can combine RL with FLC, ANN, MPC, RBC, or other control methods to improve performance and deployment relevance.	Hybrid methods increase system-design complexity and depend on suitable integration between learning and model/control components.
Multi-agent / distributed	Multi-agent RL is strongly supported for RES-integrated BEMS, districts, microgrids, and coordinated control of interacting energy assets.	MARL introduces communication, coordination, scalability, and stability challenges that still require further development.

Table S1.94: RQ1 future-direction notes for Michailidis et al. (2025), after recoding.

Survey	note_type	note
Michailidis et al. (2025)	Future direction	Actor-critic methods such as TD3, DDPG, and SAC are future-relevant for continuous and high-dimensional RES-integrated building-control problems.
Michailidis et al. (2025)	Future direction	Hybrid RL is identified as an emerging and rapidly growing direction for combining RL with established control and prediction methods.
Michailidis et al. (2025)	Future direction	Multi-agent and hierarchical approaches are important future directions for coordinated RES-integrated BEMS, districts, and microgrid-level control.
Michailidis et al. (2025)	Future direction	Future work should improve transferability, robustness, safety/compliance, privacy-preserving distributed learning, and real-world deployment.

Table S1.95: One-line RQ1 answer for Michailidis et al. (2025), after recoding.

Survey	RQ1 answer
Michailidis et al. (2025)	Michailidis et al. (2025) characterize RL for renewable-energy-integrated BEMS through a broad algorithmic taxonomy in which value-based methods remain historically important, but the strongest future-oriented recommendations shift toward actor-critic, hybrid/model-based, and multi-agent approaches for continuous, high-dimensional, coordinated, and RES-integrated building-control problems.

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